

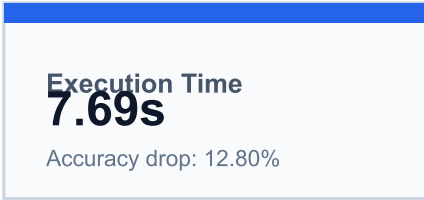
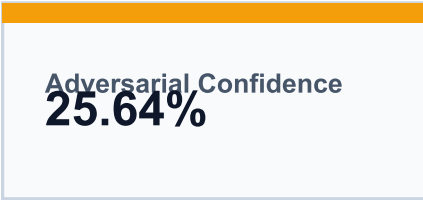
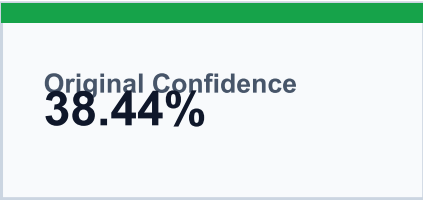
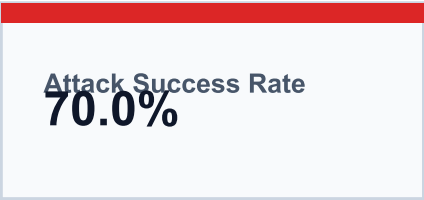
Threat Assessment Report

Executive summary for model robustness under adversarial attack

Model: microsoft/resnet-18

Attack Type: FGSM

Images Evaluated: 10



Threat Level Assessment

HIGH RISK

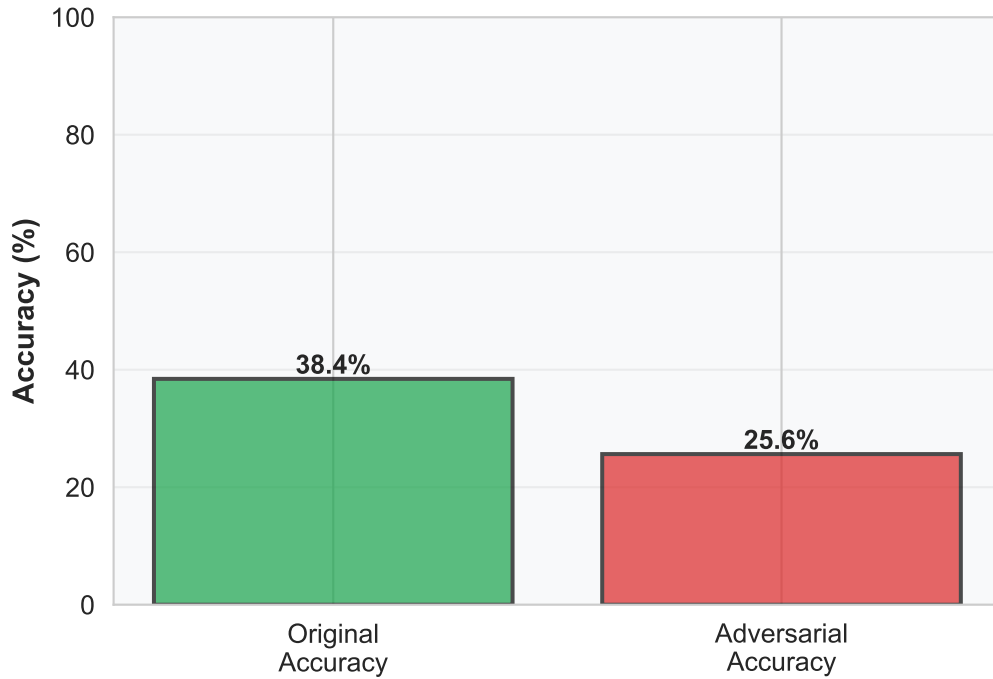
The FGSM attack altered model predictions on 70.0% of the evaluated images. Average confidence shifted from 38.44% to 25.64%, indicating the current robustness posture is high risk.

Assessment Narrative

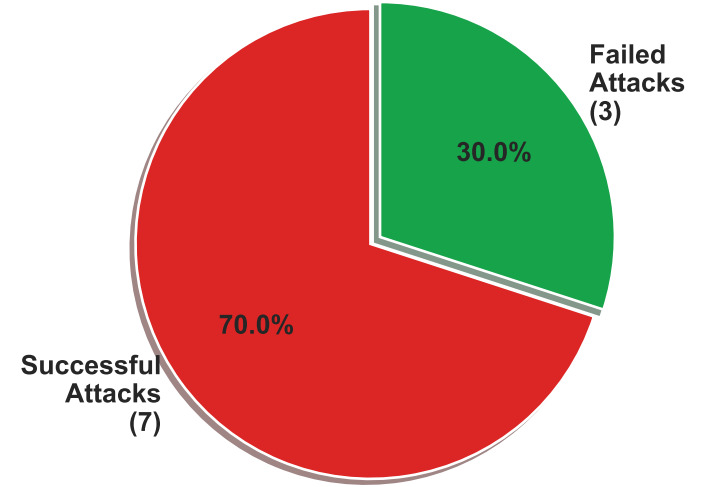
Successfully executed FGSM attack on model microsoft/resnet-18 using 10 test images. Attack success rate: 70.0%. Average original accuracy: 38.44%, Average adversarial accuracy: 25.64%. The attack successfully fooled the model in 7 out of 10 cases.

Detailed Analysis

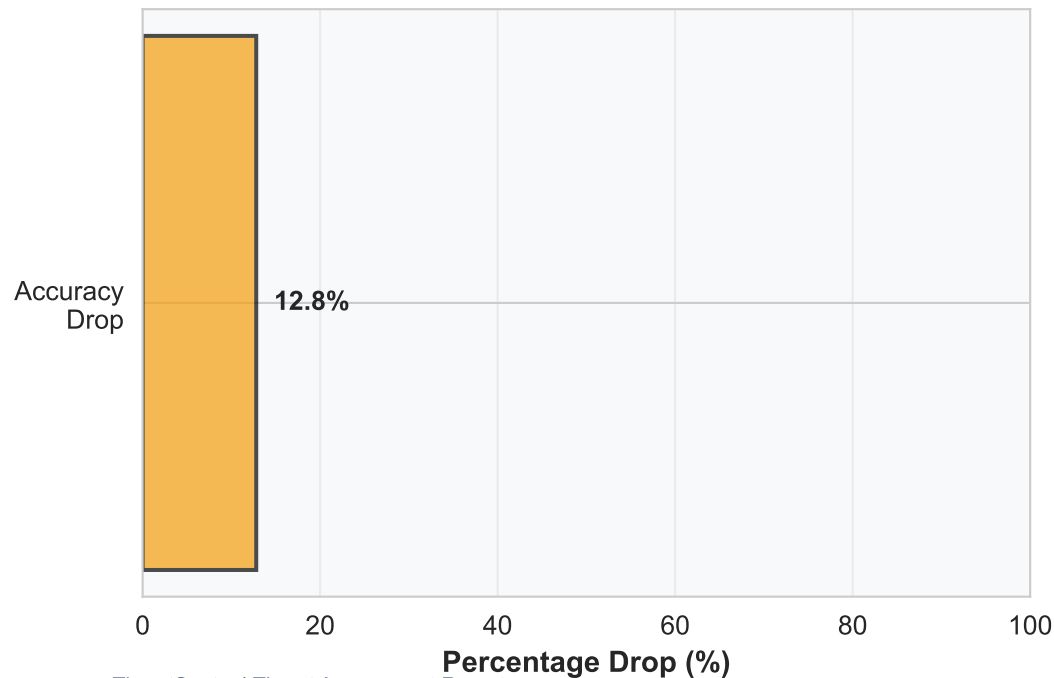
Accuracy Comparison



Attack Success Distribution



Model Robustness Impact



Confidence Comparison (Sample)

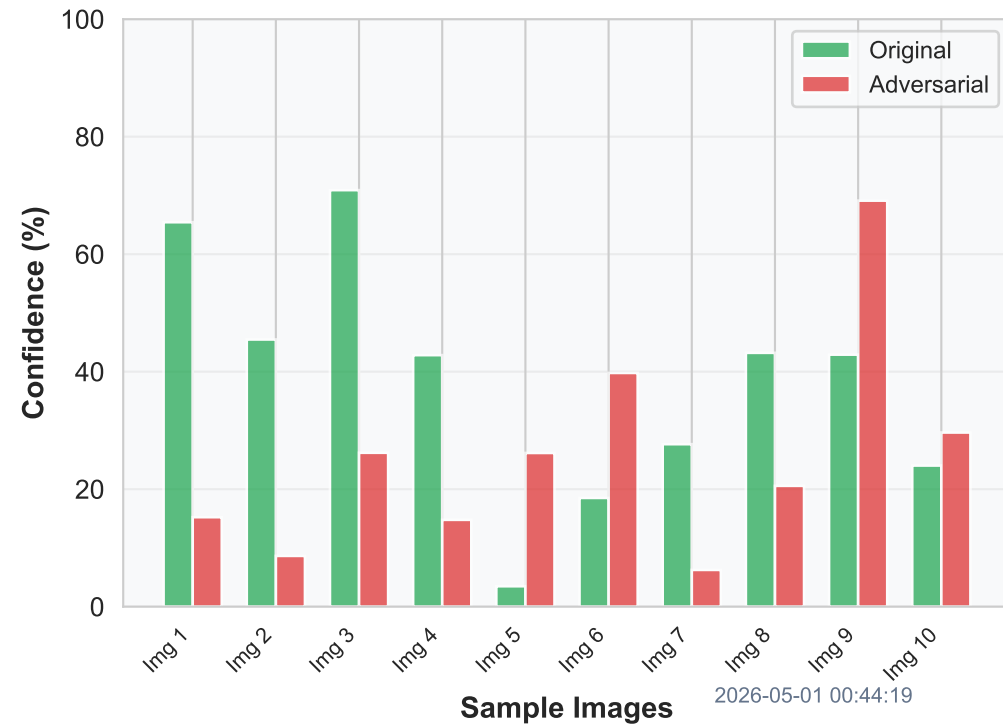
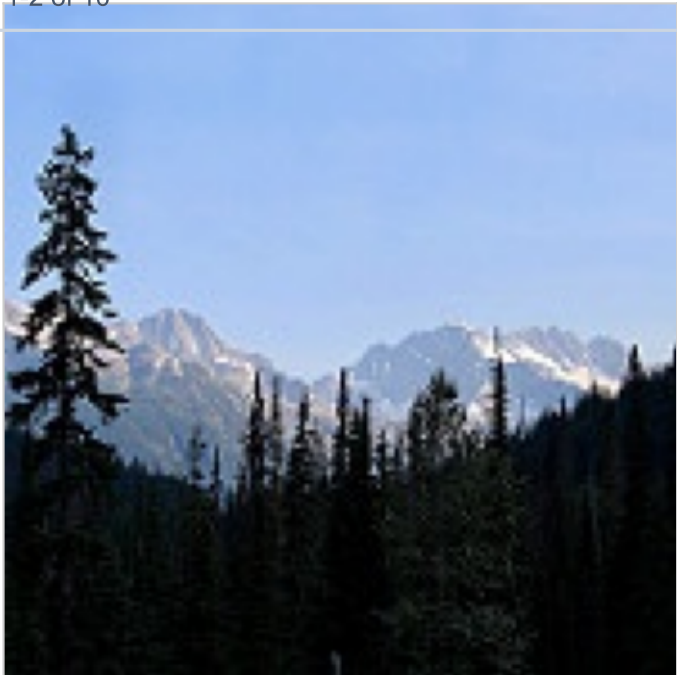


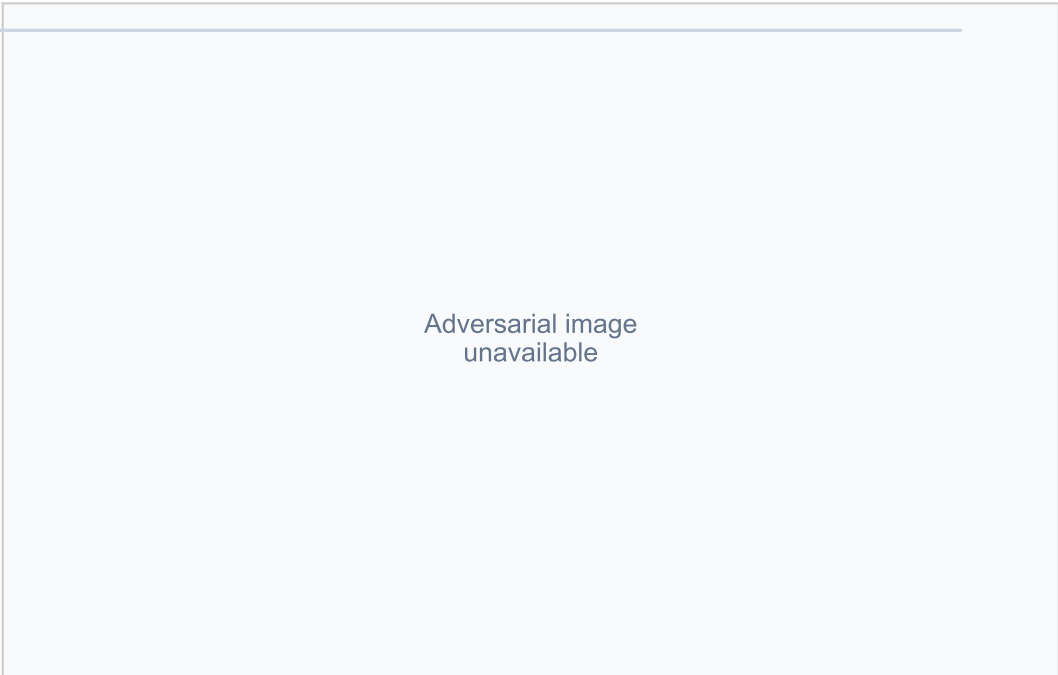
Image Evidence - Original vs Adversarial

Images 1-2 of 10

Original: alp
(70.9%)



Adversarial: alp
(26.2%)
ATTACK BLOCKED



Original: promontory, headland, head, foreland
(45.5%)



Adversarial: promontory, headland, head, foreland
(8.6%)
ATTACK BLOCKED

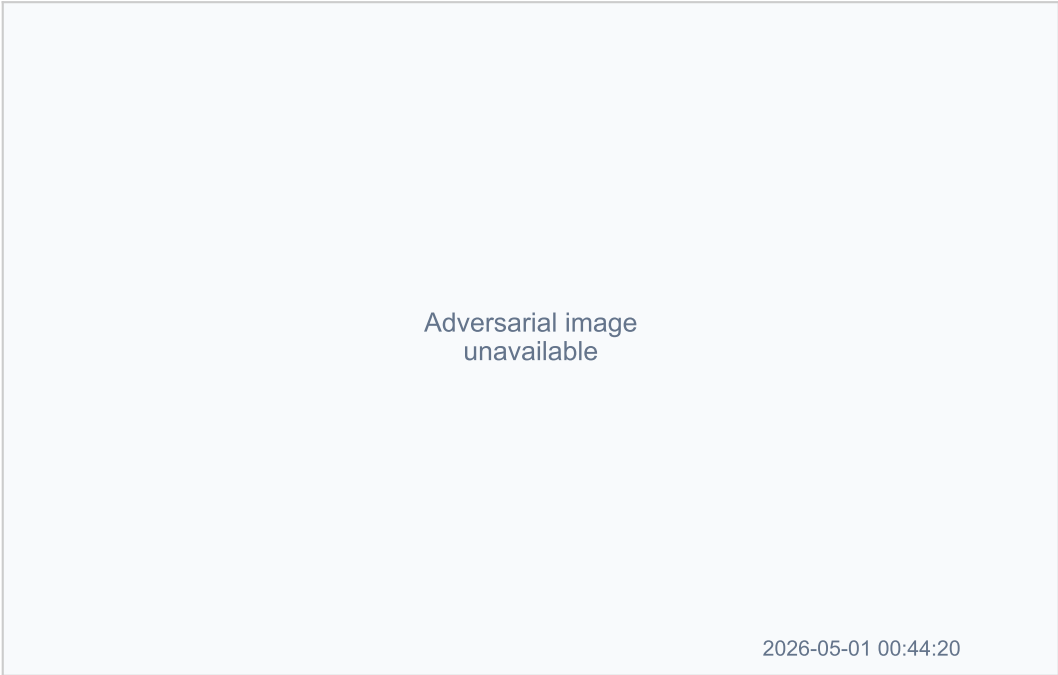
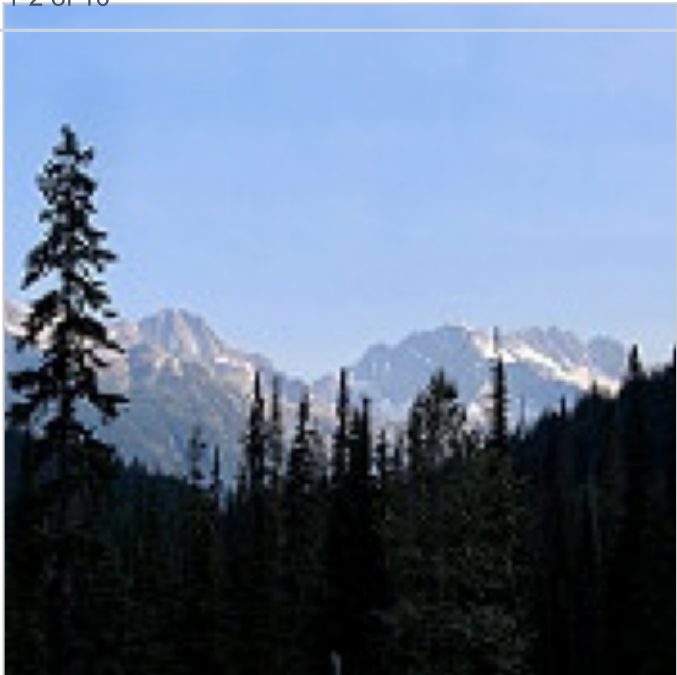


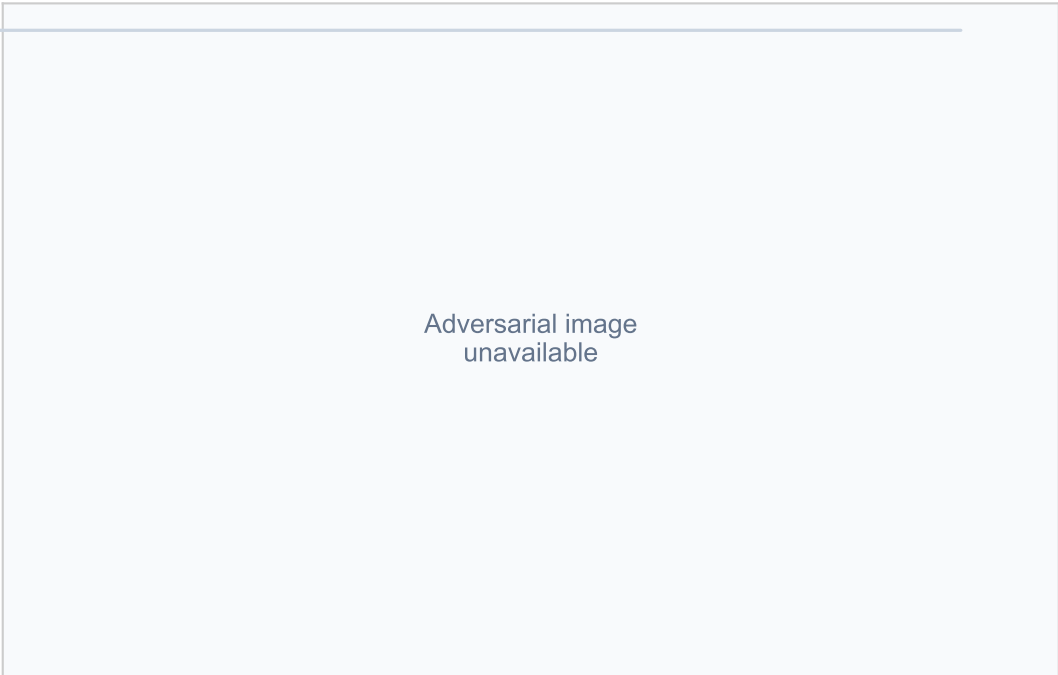
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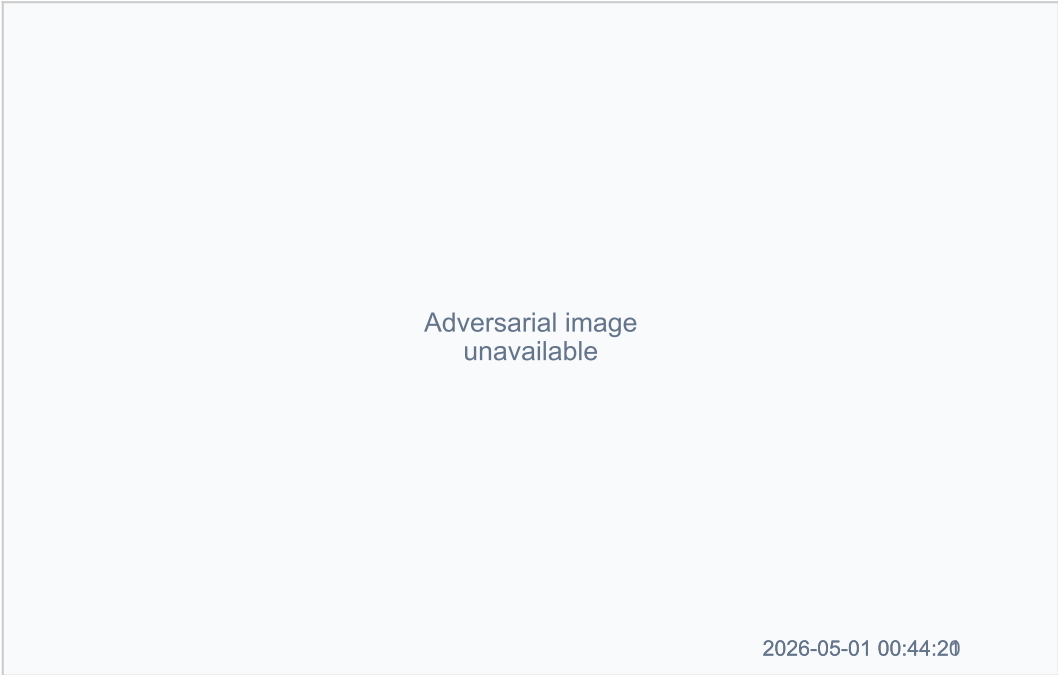


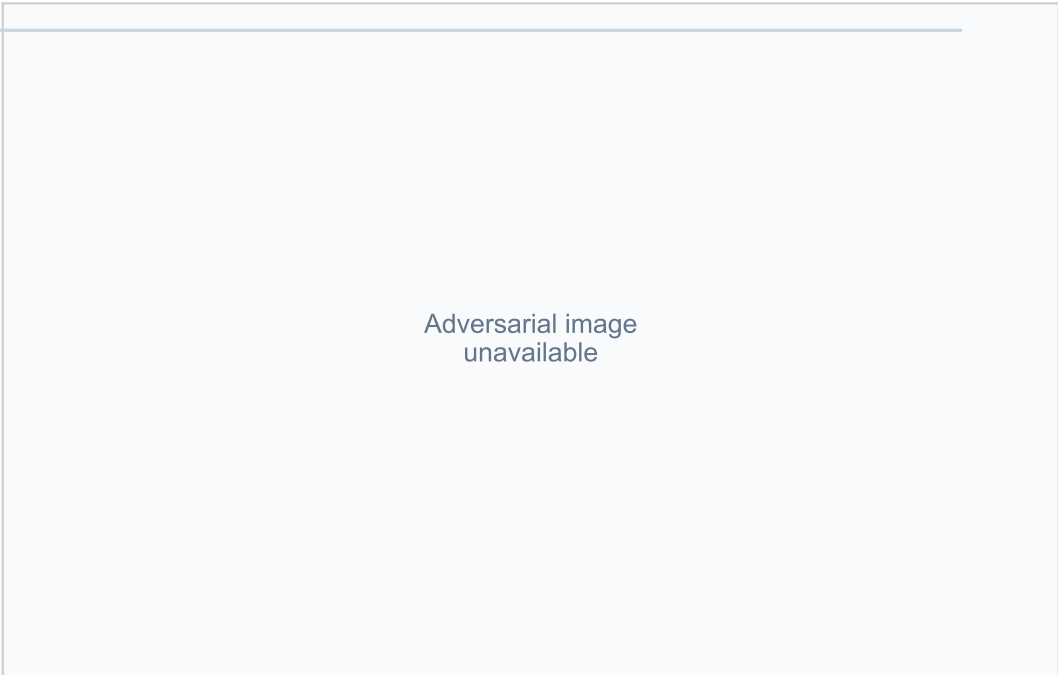
Image Evidence - Original vs Adversarial

Images 3-4 of 10

Original: viaduct
(42.8%)



Adversarial: viaduct
(14.8%)
ATTACK BLOCKED



Original: volcano
(43.2%)



Adversarial: alp
(20.6%)
ATTACK SUCCESSFUL

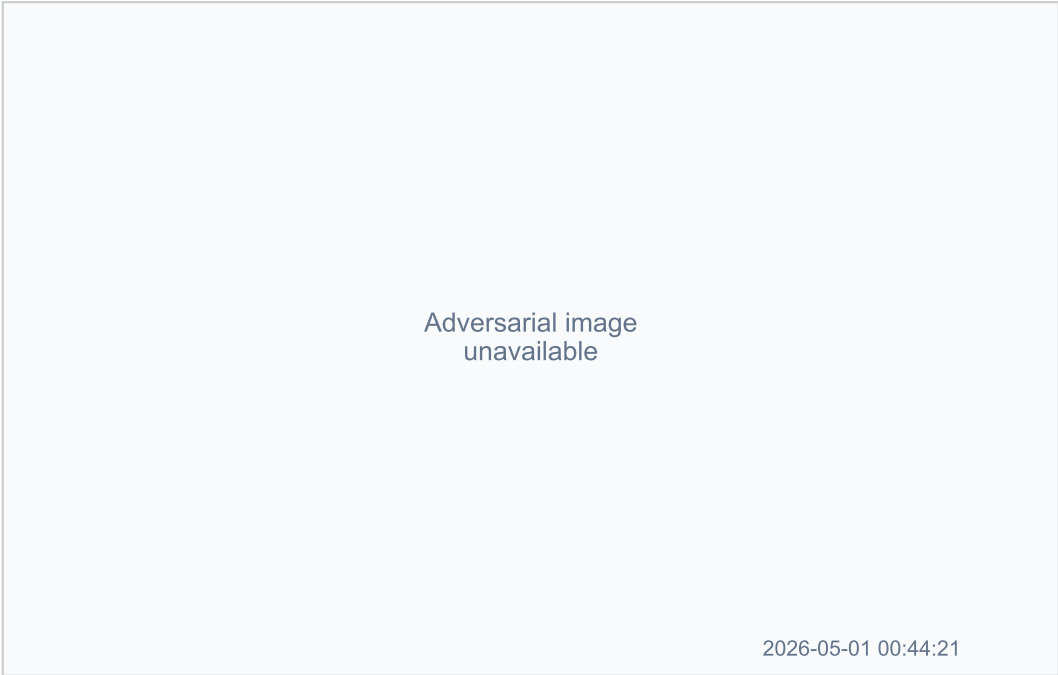


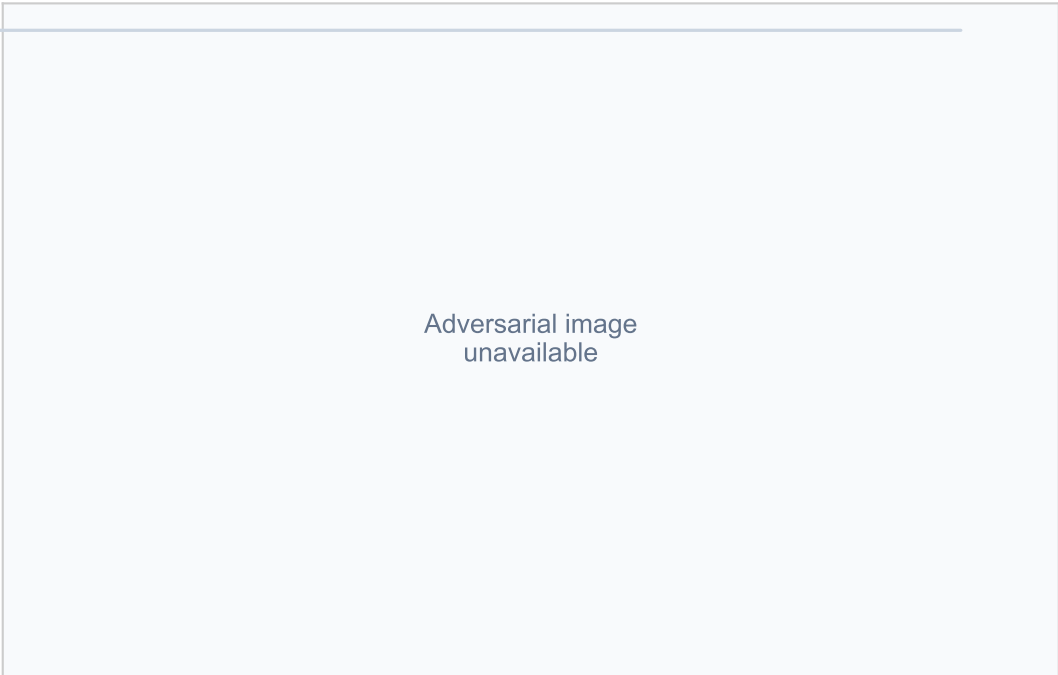
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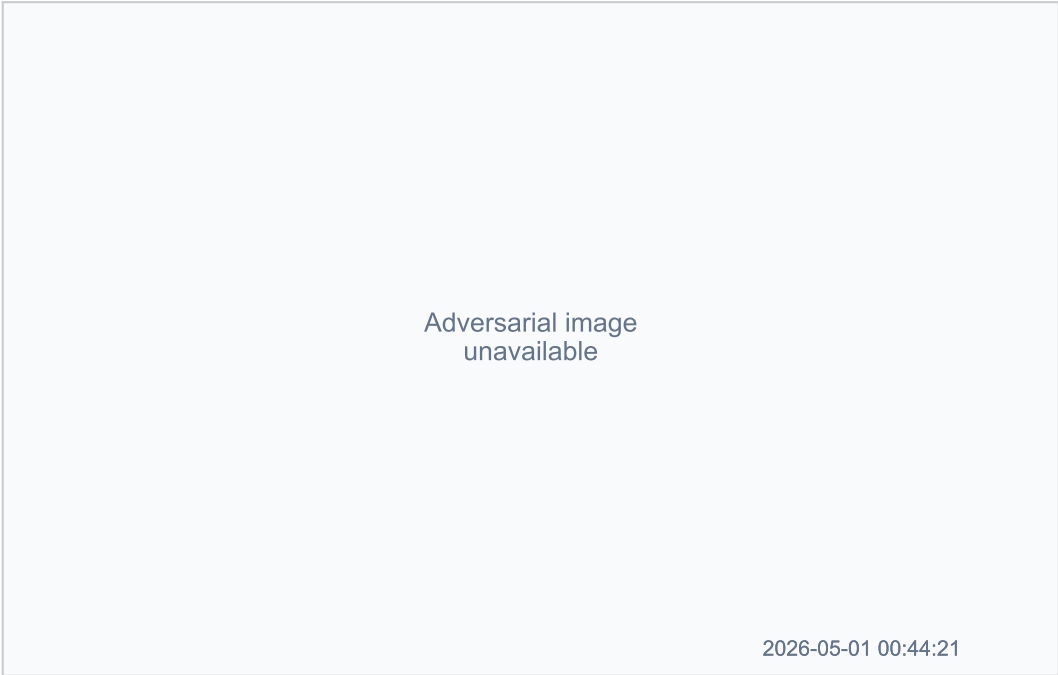


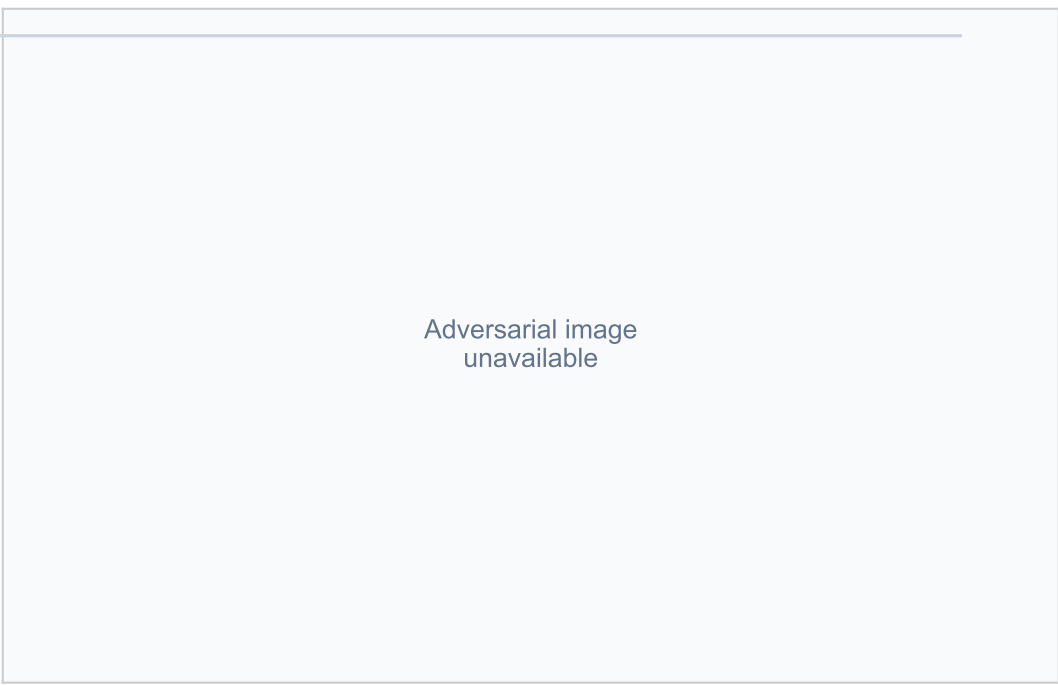
Image Evidence - Original vs Adversarial

Images 5-6 of 10

Original: monastery
(24.0%)



Adversarial: jigsaw puzzle
(29.6%)
ATTACK SUCCESSFUL



Original: lakeside, lakeshore
(27.7%)



Adversarial: alp
(6.3%)
ATTACK SUCCESSFUL

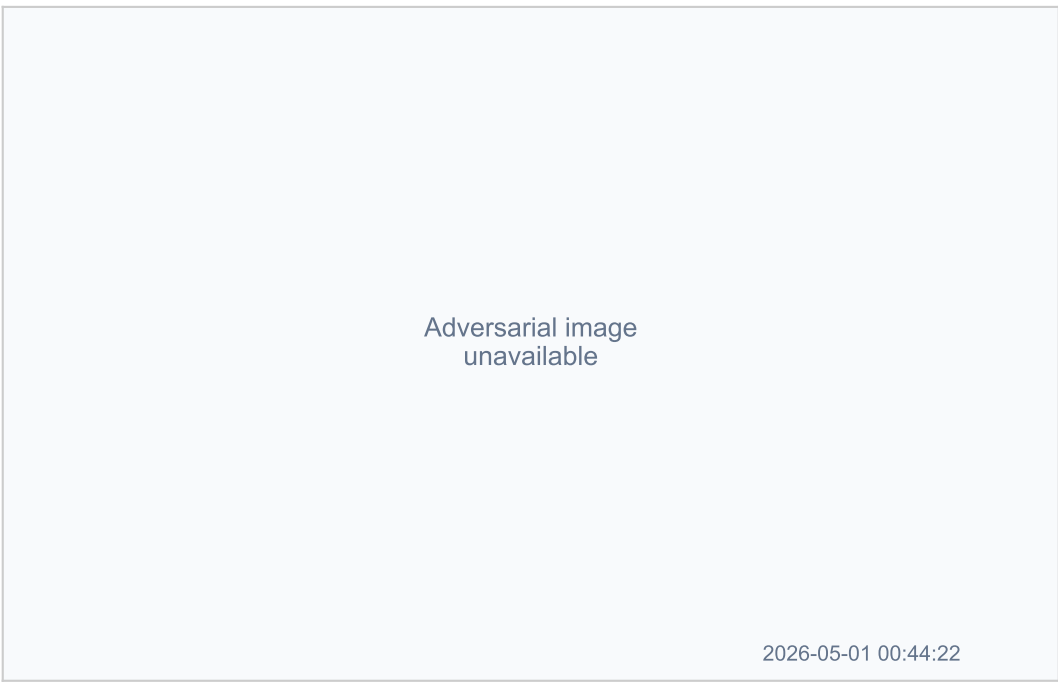


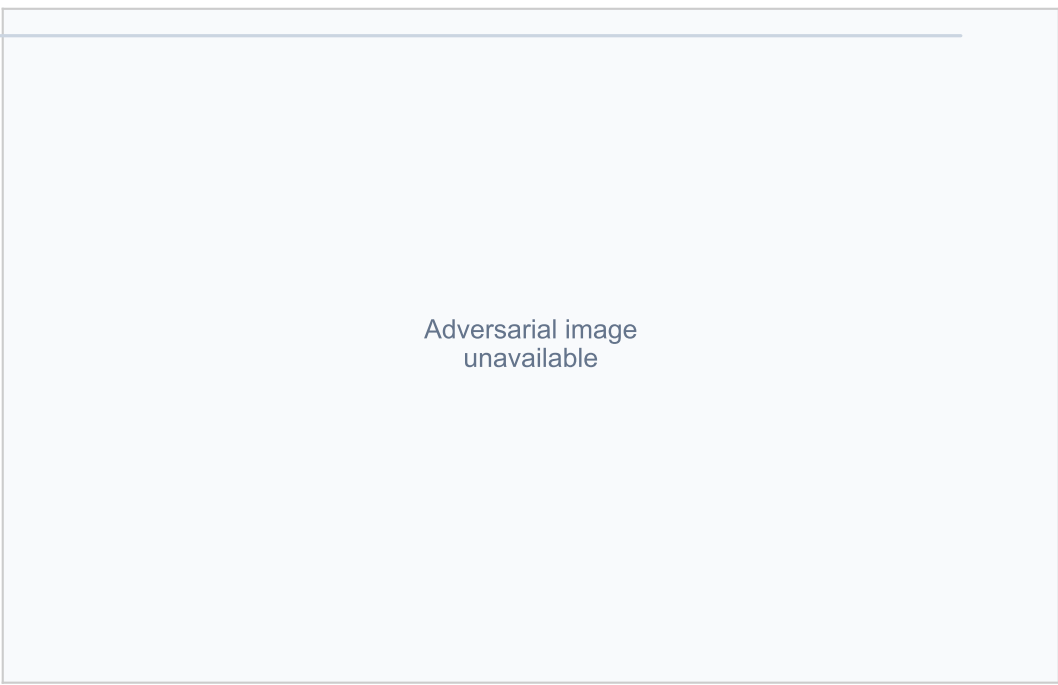
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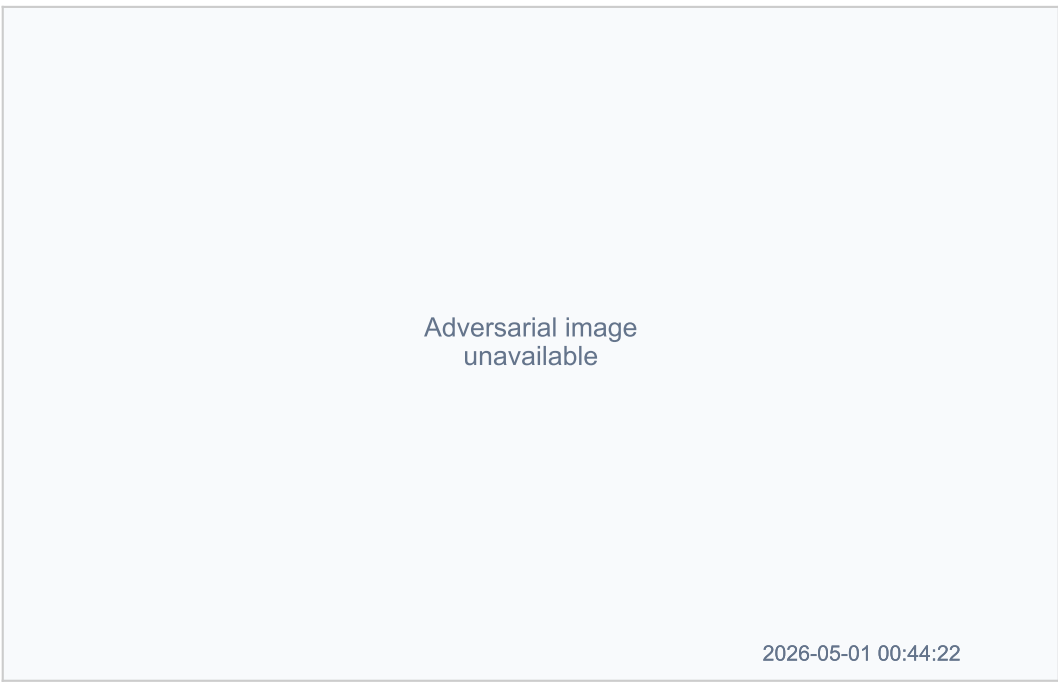


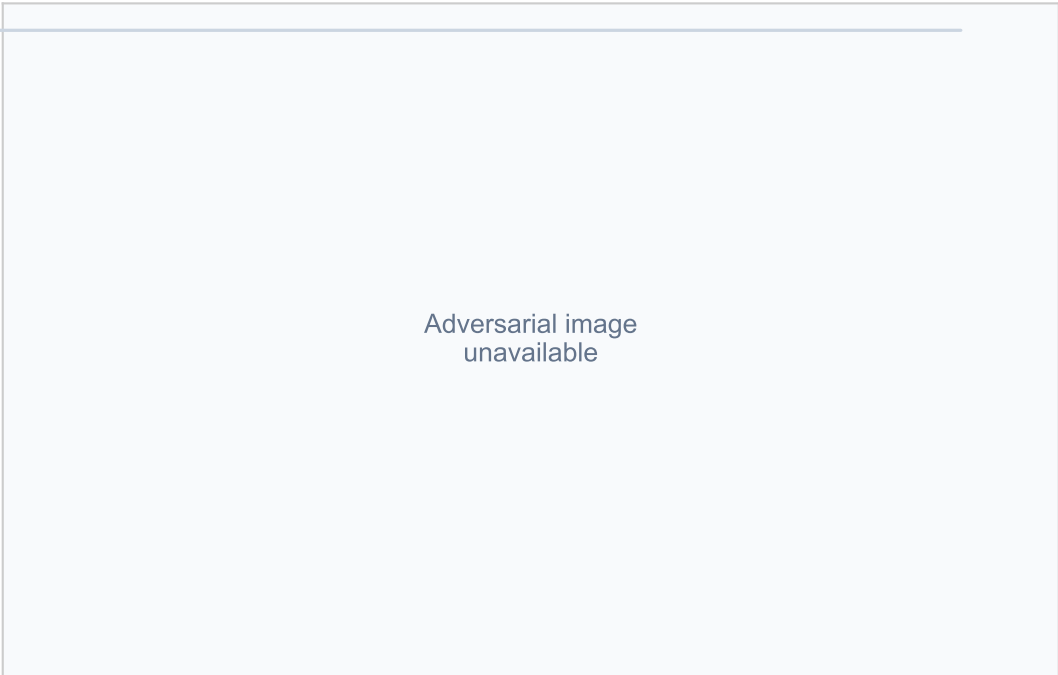
Image Evidence - Original vs Adversarial

Images 7-8 of 10

Original: alp
(65.4%)



Adversarial: volcano
(15.2%)
ATTACK SUCCESSFUL



Original: eel
(42.9%)



Adversarial: lacewing, lacewing fly
(69.1%)
ATTACK SUCCESSFUL

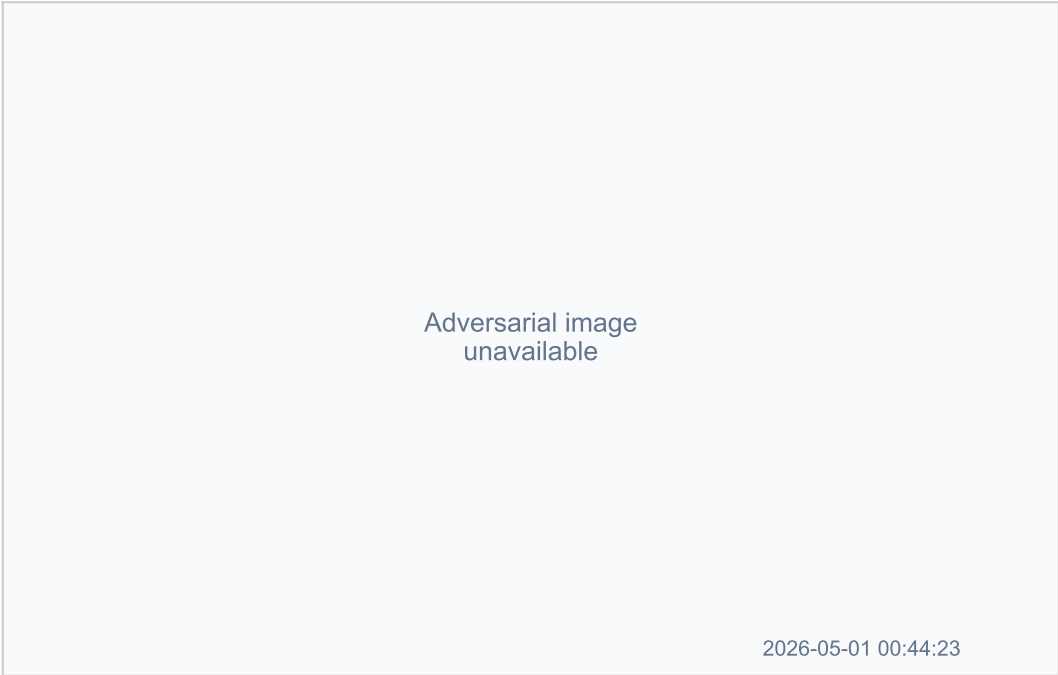


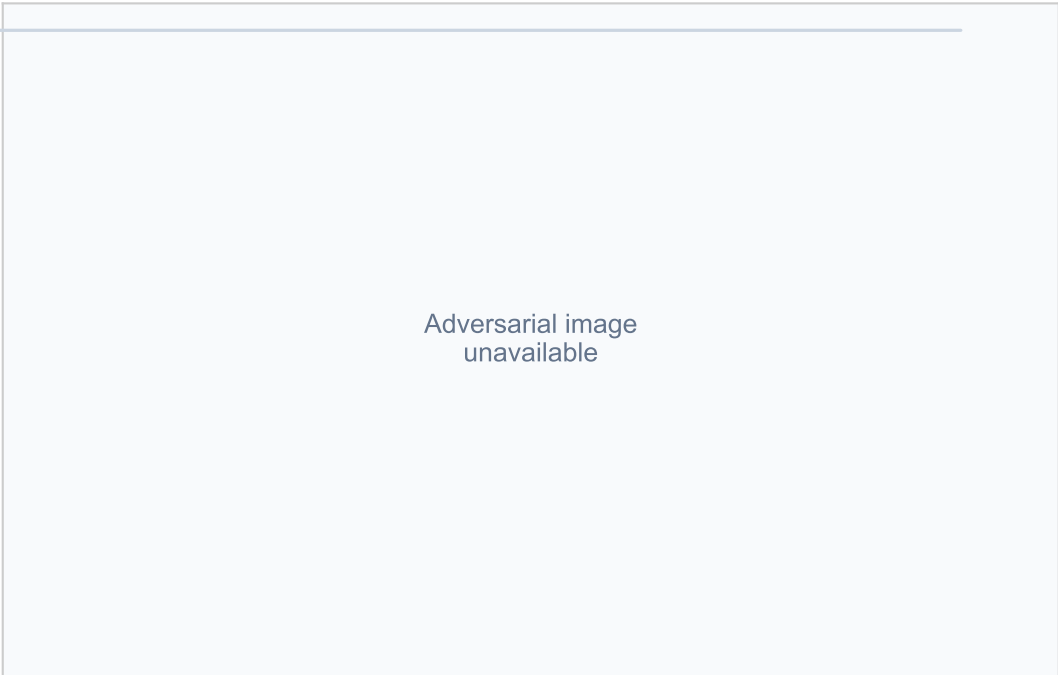
Image Evidence - Original vs Adversarial

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(65.4%)



Adversarial: volcano
(15.2%)
ATTACK SUCCESSFUL

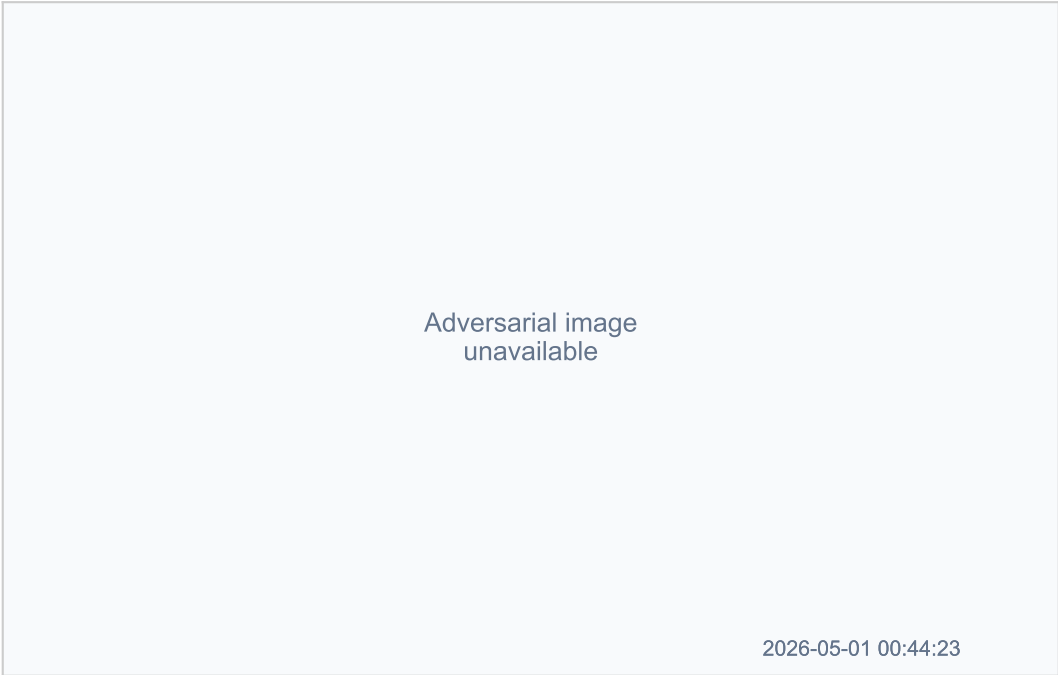


Adversarial image
unavailable

Original: eel
(42.9%)



Adversarial: lacewing, lacewing fly
(69.1%)
ATTACK SUCCESSFUL

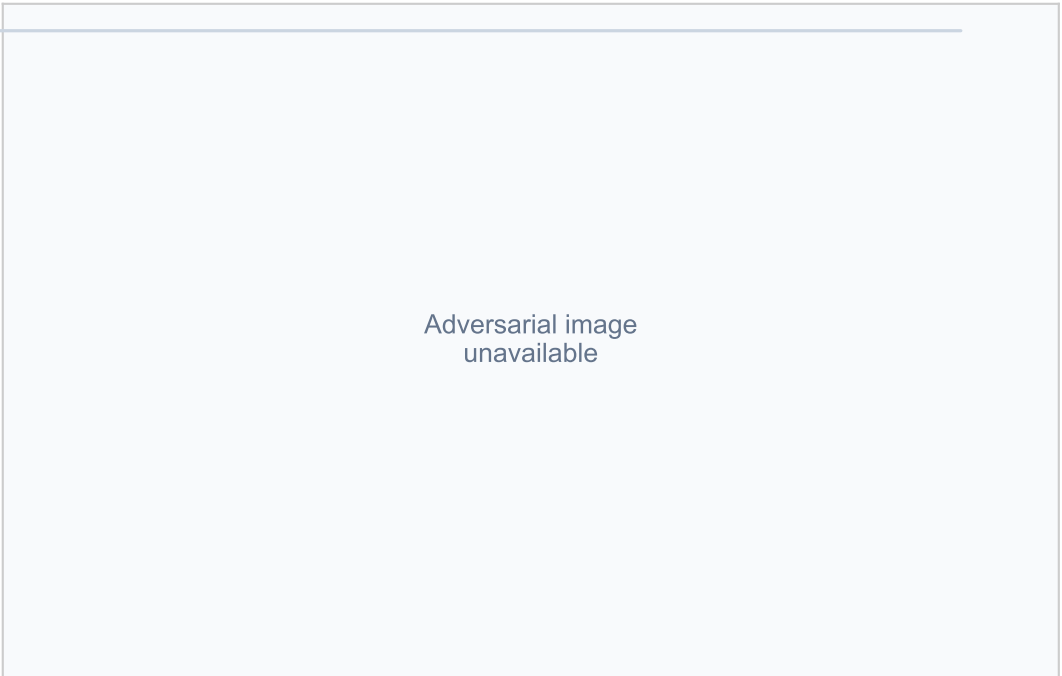


Adversarial image
unavailable

Image Evidence - Original vs Adversarial

Original: cinema, movie theater, movie house, picture palace
(18.5%)

Images 9-10 of 10



Adversarial: gondola
(39.8%)
ATTACK SUCCESSFUL

Adversarial image
unavailable

Original: maze, labyrinth
(3.5%)



Adversarial: American alligator, Alligator mississippiensis
(26.2%)
ATTACK SUCCESSFUL

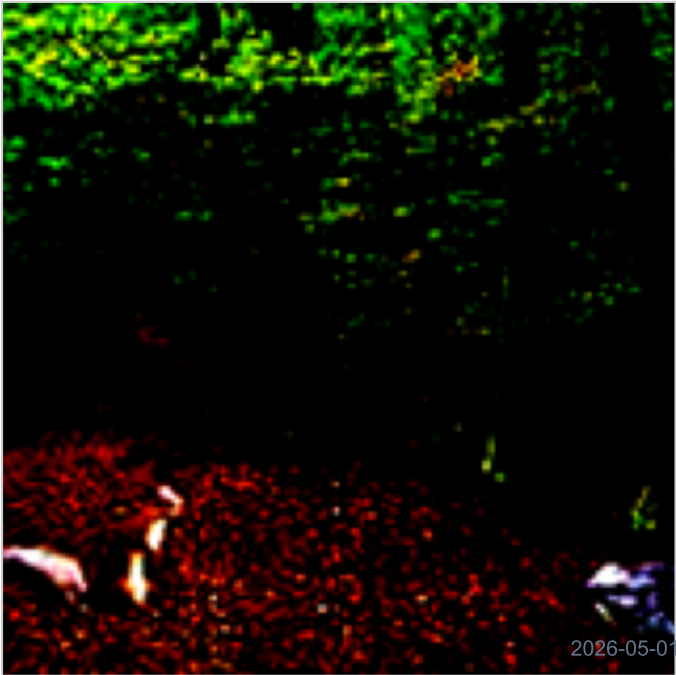


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Original: cinema, movie theater, movie houses, picture palace
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Images 9-10 of 10



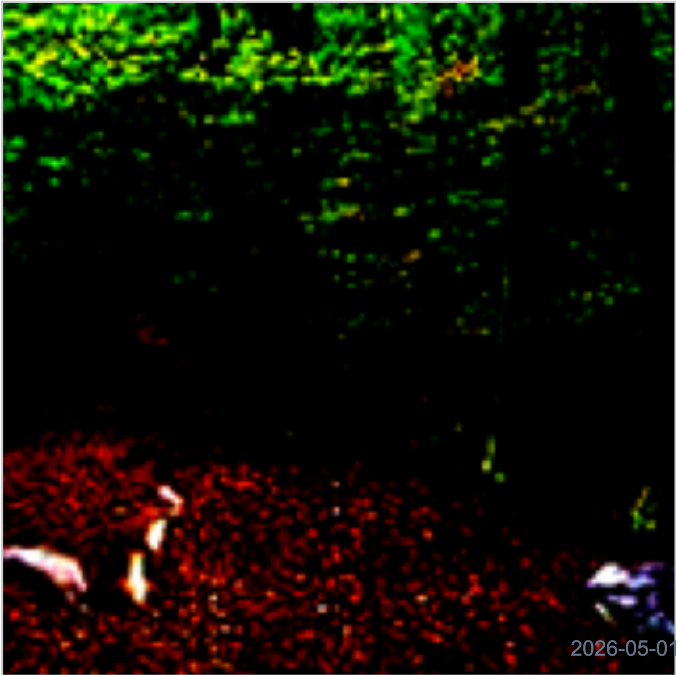
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Adversarial image
unavailable

Original: maze, labyrinth
(3.5%)



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ATTACK SUCCESSFUL



Recommendations

Security actions based on observed threat exposure

Assessment Summary

Successfully executed FGSM attack on model microsoft/resnet-18 using 10 test images. Attack success rate: 70.0%. Average original accuracy: 38.44%, Average adversarial accuracy: 25.64%. The attack successfully fooled the model in 7 out of 10 cases.

Security Recommendations

1. Implement Adversarial Training

- Retrain your model with adversarial examples to improve robustness
- Use techniques like FGSM, PGD during training phase

2. Add Input Validation & Preprocessing

- Implement input sanitization and anomaly detection
- Use defensive distillation or feature squeezing

3. Deploy Ensemble Methods

- Use multiple models with different architectures
- Implement voting mechanisms for predictions

4. Continuous Monitoring

- Set up real-time performance monitoring
- Detect and alert on unusual prediction patterns

5. Regular Security Audits

- Conduct periodic threat assessments
- Stay updated with latest attack techniques